

Nancy Jain

GenAI / LLM / Agentic AI Engineer

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PROFESSIONAL SUMMARY

GenAI / LLM Engineer with nearly 3 years shipping production LLM systems and agents – RAG pipelines, multi-step orchestration, tool / function calling, and self-correcting agentic workflows. At DataWeave, built **AskAI** (LangGraph agentic text-to-SQL), **enterprise attribute tagging** (NER, classifiers, **28+** types), **MetaMapper** (**2M+** records, graph retrieval + reranking), and **semantic product matching** (hybrid retrieval + reranking). Also delivered RAG products with LangChain, FastAPI, and vector databases at Corazor and on a live community platform.

TECHNICAL SKILLS

Agentic AI: AI Agents, Agentic Workflows, LangGraph, Multi-Step Task Planning, Tool / Function Calling, MCP (Model Context Protocol), LLM Orchestration, Self-Healing / Reflection Loops, Conversational Memory & State Management, LLM-as-Judge Evaluation, Text-to-SQL (NL2SQL), LiteLLM

GenAI & LLM: Retrieval-Augmented Generation (RAG), Semantic Search, Embeddings, Top-K Retrieval, Vector Databases (Qdrant, ChromaDB), Cross-Encoder Reranking, Solr, NER, Text Classification, Multimodal LLMs (VLM), Prompt Engineering, Fine-Tuning, LLM Evaluation, NLP, LangChain, OpenAI API, Gemini API, AWS Bedrock (Claude), Hugging Face

Backend & MLOps: Python, FastAPI, REST APIs, Server-Sent Events (SSE), Async Inference, Docker, Kubernetes, CI/CD, Git, Kafka, Prometheus, MLflow, Event-Driven Microservices

Data & Cloud: SQL, Pandas, NumPy, PostgreSQL, Redis, MongoDB, DuckDB, Snowflake; AWS – Bedrock, Athena, S3, SQS, STS, ECR

PROFESSIONAL EXPERIENCE

Data Scientist (GenAI / LLM & Agentic AI)

Mar 2024 – Present

DataWeave Inc.

- Built **AskAI**, a production **agentic text-to-SQL system** on **LangGraph** (13+ node StateGraph, FastAPI, AWS Bedrock Claude) that autonomously plans, generates, and executes SQL over **Athena/DuckDB** via **conditional routing**, **RAG** schema grounding (ChromaDB + Titan embeddings), and **tool-calling** KPI computation.
- Engineered its **self-correction loops** – an **LLM-as-judge** validator that semantically verifies results and triggers SQL regeneration, plus bounded execution-error retries – and **cross-turn conversational memory** resolving follow-ups in-memory (sub-second, no LLM/DB call) with **Redis** result externalization + **Postgres** checkpointing.
- Built **enterprise attribute tagging** (**FastAPI** + **Celery**, **K8s/SQS**; NER, classifiers, **SDE**, KG normalization, **LLM fallback**; **28+** types).
- Built **MetaMapper**, classifying **2M+ product records** into canonical retail taxonomies (embeddings → Qdrant → Bedrock Claude, **80% to 97%** accuracy), and engineered a **graph-based retrieval + reranking** engine (NetworkX hierarchy graph, hybrid vector + BM25, cross-encoder rerank, confidence routing) that lifted long-tail coverage $\sim 3\times$ (**94.7% Recall@5**) and cut LLM calls **30–40%**.
- Developed **semantic product matching** (million-scale Solr): **BM25** + **BGE-M3** + **RRF**, **HNSW**, cross-encoder rerank, **LLM validation**, Exact/Similar/Reject (**Top-1**, **MRR**, **NDCG**, **AUC**); **Image Similarity API** (Bedrock).

AI Engineer

Oct 2023 – Mar 2024

Corazor Technology Private Limited

- Built **SpectrAble**, a **RAG-based insight engine** with a **feedback-driven re-ranking loop** (human-in-the-loop) where clinician validations dynamically adjusted retrieval ranking using relevance scoring, improving grounding over baseline.
- Delivered an **event-driven reconciliation system** with **28+ anomaly-detection use cases** and a **Gemini/Dialogflow conversational agent**, deployed via an async Node-Python inference bridge (caching, fallback) under **DPDP/GDPR** compliance.

PROJECTS

Outbound Cockpit — LangGraph + MCP Sales Research

[Live](#) | [GitHub](#)

- LangGraph** orchestrator with **12 FastMCP tools**, **LLM-as-judge validation loops**, and **MongoDB thread memory** – replacing a monolithic prompt→JSON pipeline.

AI Integration – Execute Community (EXCOM)

[Live](#) | [GitHub](#)

- RAG-powered assistant** on a live community platform with **MLflow** prompt evaluation and AI moderation (NLP similarity scoring).

EDUCATION

B.Tech in Computer Science and Engineering

Dr. A.P.J. Abdul Kalam Technical University, India

Jul 2020 – Jul 2024

CGPA: 7.0/10.0

ACHIEVEMENTS & CERTIFICATIONS

- 94.38 Percentile (Rank 275 / 4,892)** – Amazon ML Challenge: Product Length Prediction (Apr 2023)
- 2nd Runner-Up** – MAMC Health-A-Thon: Depression Detection (Oct 2022)
- Machine Learning Certification** – IBM SkillsBuild (May 2023) [\[Verify\]](#)